Design and implementation of a Database of the 2025 Formula 1 Season



Anna Rebecca Sala - MAT. 894569
Giorgia Brioschi - MAT. 894407

Data Management Project
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Introduction

The aim of this project is to design and build a comprehensive database of the 2025 Formula 1 season to support analytical queries and answer research questions

Project Goal

Provide a complete data source enabling analysis of races, results, drivers, constructors, circuits, and standings.

Support research questions that explore relationships, trends, and insights within the 2025 season.

Phases:

1. Data Acquisition: API and scraping
2. Data Preparation: Data Cleaning, Data Integration and Enrichment, Data Quality Verification
3. Data Storage: Database population and Queries implementation

Data Acquisition

Collecting and importing data to be used in subsequent stages of the workflow, relevant to the project objectives

Data Sources:

1. Jolpica F1 API: Provides structured JSON data on races, sprints, drivers, constructors, circuits, and standings. Selected for its completeness and coverage of key F1 entities.
2. Wikipedia: Hosts multiple pages with detailed, structured tables on circuits, drivers, and constructors, including historical and descriptive information.

Techniques Used:

1. API Requests: filtered endpoints to select only relevant 2025 season data.
2. Web Scraping: automated extraction and parsing of HTML tables, filtering and cleaning records.
3. Execution: All performed in Python using Jupyter Notebooks.

Output: *Raw data stored as CSV files, ready for data cleaning and preprocessing.*

Data Integration and Enrichment

Combining the two data sources by establishing connections between corresponding datasets

Integration Strategy:

- Drivers: surname - full name
- Constructors: alphabetical sorting
- Circuits - Races: matched via Grand Prix name
- Results + Sprints: concatenated into a single dataset with an isSprint flag

Characteristics:

- Fully automated using Python functions
- High matching rate between API and scraped datasets
- No incorrect matches after validation
- Achieved thanks to prior data cleaning and preprocessing, which ensured complete, consistent, and non-redundant datasets

Output: *Integrated CSV files.*

Data Quality

Improving quality, clarity, and overall consistency of the datasets

- ***Completeness:*** *presence of all and only the aspects of reality relevant to the project.*

Issue: Single sources were missing complementary information.

Solution: Integration of multiple sources ensured a complete dataset with all relevant attributes, .

- ***Consistency:*** *whether data is recorded or collected in a uniform way.*

Issue: Some attributes had different formats, special characters, or inconsistent types.

Solution: Removed irrelevant records, standardized text and data types, filtered to 2025 season, adjusted formats to match data types.

- ***Redundancy:*** *presence of duplicate or unnecessary information.*

Issue: Some tables contained overlapping attributes for drivers and constructors.

Solution: Redundant columns were removed.

Output: *Cleaned and processed CSV files ready for data integration.*

Data Storage

Store the integrated data in a flexible database

Database Choice: *MongoDB, a document-based NoSQL database*

Design Considerations:

- Greater schema flexibility than relational databases.
- Supports queries across collections with a large amount of attributes.
- References used to avoid redundancy (drivers, constructors, circuits linked to results via IDs).

Database Structure:

- Database f1_2025 contains five collections: drivers, constructors, circuits, races, results.
- Collections linked through identifiers:
  - driverId → results and drivers
  - constructorId → results and constructors
  - circuitId → races and circuits
  - round → races and results

Output: *Fully integrated database for detailed analysis of the 2025 F1 season.*

Queries

Queries & Insights:

- *Are there performance differences between races held on street circuits versus permanent circuits?*

The query calculates the percentage of DNF by drivers out of the total races, grouping race results by circuit type, by integrating race outcomes with circuit data.

Output: Higher DNF rate on street circuits (14.44%) vs race circuits (10.98%).

- *Does grid position significantly influence a driver's final race position?*

The query quantifies the position gain or loss between the grid position and the final race position, grouped by grid position.

Output: Front positions (1–5) show small position changes; midfield (6–12) moderate; back positions (13–20) can gain/lose up to 5 places.

- *What are the main characteristics and 2025 performance metrics of a driver?*

The query aggregates personal information, seasonal results and historical performance for a selected driver, by combining multiple collections through join operations.

Output: Charles Leclerc – 30 races, 242 points, 7 podiums in 2025.

Conclusions

Key Achievements:

- Developed a comprehensive and integrated database of the 2025 F1 season.
- Successfully cleaned, standardized, and merged heterogeneous data sources (API + Wikipedia).
- Database supports complex queries combining drivers, constructors, races, circuits, and results.

Challenges & Solutions:

- Handled missing or inconsistent records from multiple sources.
- Removed redundant attributes and standardized data formats.

Future Developments:

- Integration of additional seasons for historical comparisons.
- Keeping the database up-to-date with ongoing F1 events.
- Foundation for predictive models and further advanced analytics.